

Senior Project Proposal

Project Name:

- *EscapeAssist* – An AI-powered Ford Escape Owner's Manual Assistant

Team and Contact Information:

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Stakeholders:

- **Primary Stakeholders:**
 - Ford Escape Owners (2022 model year) – Individuals who need quick, accurate answers from their owner's manual.
 - Nicholas (Developer) – Responsible for design, implementation, evaluation, and delivery.
- **Secondary Stakeholders:**
 - Future/Past Escape Owners (2001 – 2021, 2023 - 2026) – The architecture can be expanded to support additional model years.
 - Automotive Service Advisors / Dealership Staff – Frequently answer repetitive questions that the assistant could help automate.
 - Individuals with Accessibility Needs – Users who struggle with long technical documentation benefit from simplified, natural-language answers.

Project Purpose:

Owner's Manuals are long, difficult to search through quickly, and can be very confusing. The 2022 Ford Escape manual is over 500 pages, and finding information quickly – such as information about warning lights, towing limits, hybrid behavior, oil specifications, etc. – is challenging for many drivers, even those who are familiar with navigating Owner's Manuals.

EscapeAssist solves this problem by giving Escape owners a natural-language assistant that retrieves accurate, grounded answers from the manual nearly instantly. This will improve accessibility, safety, vehicle care, and owner confidence.

Background / Prior Knowledge:

Through my coursework I have gained experience with Python, APIs, the Software Development Life Cycle, Software Engineering principles, Full-Stack Web Development, and Testing. I also have exposure to LLMs and experience using AI tools.

- **Prior Work by Others:**

- RAG (Retrieval Augmented Generation) systems, vector databases, and LLM-powered assistants already exist, but none that I personally know of are tailored to vehicle manuals or the Ford Escape specifically. *EscapeAssist* builds on all these concepts but applies them to a new domain. Microsoft Foundry provides a built-in RAG pipeline that I can build on.
- **Prior Work by Me:**
 - I have experimented with LLMs and AI tools, but I have never built a RAG agent, full evaluation/testing harness, or dashboard before. This project is a significant expansion beyond anything I have done previously.
- **Foundational Course Background:**
 - My chosen degree emphasis was Full-Stack Web Development, which has given me foundational knowledge in building backend systems, front-end UIs, structured applications, and testing strategies. This project will extend those skills into modern AI engineering.
 - I consider myself a newbie in RAG systems, LLM evaluation, and Microsoft Foundry.

Project Description:

- **Detailed Project Explanation:**
 - *EscapeAssist* will:
 - Allow users to ask natural-language questions about their Ford Escape.
 - Retrieve relevant sections from the 2022 owner's manual.
 - Provide powertrain-specific answers (Hybrid, PHEV, EcoBoost, AWD/FWD).
 - Provide accurate information from the 2022 Ford Escape manual using preprocessing-optimized documents.
 - Include an automated evaluation harness using a judge LLM.
 - Include a dashboard showing accuracy, hallucination rate, and performance metrics.
- **Existing Solutions:**
 - Ford provides owners with access to a digital manual (HTML and PDF), but it is not conversational, not context-aware, and not optimized for quick retrieval. *EscapeAssist* is more intuitive, faster, and more accessible than the manual itself.
 - Existing LLM Chatbots can answer questions based on an internet search, but they are prone to answering based on misinformation and hallucinations.
- **Real-World Impact:**
 - This tool helps vehicle owners quickly access critical information, which will improve safety and reduce owner frustration. It also provides increased support to users who struggle with technical documentation.
- **Intended Audience:**

- Primary: 2022 Ford Escape Owners
- Secondary: Future/Past Escape owners (2001-2021, 2023-2026), dealership staff, and individuals with accessibility needs
- **Geographic Scope:**
 - Usable anywhere, since it is a digital AI assistant.
- **Completion Criteria:**
 - This project is complete when:
 - The Foundry agent retrieves accurate information from the 2022 Ford Escape manual using preprocessing-optimized documents.
 - The Streamlit UI provides an end-to-end conversational interface that successfully queries the agent.
 - A golden question dataset is created covering major categories (warnings, towing, hybrid behavior, safety, maintenance, etc.).
 - An automated evaluation harness runs the entire dataset, collects model answers, and sends them to a judge LLM.
 - The judge LLM scores accuracy and hallucinations, producing structured evaluation results.
 - A dashboard visualizes evaluation metrics, including accuracy, hallucination rate, and category-level performance.
 - The system can be demonstrated end-to-end to the instructor, showing retrieval, evaluation, and analysis.

Significance:

EscapeAssist meets the expectations for a significant senior project because it demonstrates real AI engineering, evaluation methodology, and full-stack development.

- **Real World Impact:**
 - EscapeAssist helps vehicle owners quickly access critical information from a 500-page manual, improving safety, accessibility, and user confidence.
- **Resume-Worthy:**
 - The project showcases modern AI engineering skills, including retrieval-augmented generation (RAG), document preprocessing, LLM evaluation, prompt engineering, and dashboard analytics.
- **Learn Something New:**
 - The project requires learning how to design an evaluation harness, build a judge LLM, perform preprocessing-based RAG tuning, and analyze retrieval performance.
- **Professional Quality:**
 - The final system includes a working agent, a UI, an automated evaluation pipeline, and a dashboard, all of which demonstrate practical engineering ability.

- **Expandable:**
 - The architecture can support additional model years, additional vehicle models, or even other domains beyond automotive manuals.
- **Resume Line Example:**
 - “Designed and built an AI-powered retrieval-augmented assistant using Microsoft Foundry, Streamlit, and a custom LLM evaluation/testing harness to provide accurate, context-aware answers from a 500-page technical manual.”

New Computer Science Concepts:

To complete this project, I will learn several modern AI engineering and software engineering concepts not covered in previous coursework:

- **Document Preprocessing for RAG:**
 - Splitting, cleaning, and structuring large technical documents to improve retrieval quality.
- **Prompt Engineering for Grounded Retrieval:**
 - Designing system prompts and user-query rewriting strategies to reduce hallucinations.
- **LLM-Based Evaluation (Judge Models):**
 - Using a secondary LLM to score accuracy, detect hallucinations, and evaluate grounding.
- **Golden-Set Testing:**
 - Creating a structured dataset of questions and expected answers to measure system performance.
- **Evaluation Harness Design:**
 - Automating the process of querying the agent, collecting responses, and scoring results.
- **Dashboard Analytics:**
 - Visualizing accuracy, hallucination rate, and category-level performance using Python and Streamlit.
- **Error Analysis:**
 - Identifying failure modes and iterating on preprocessing and prompts to improve retrieval.

These concepts extend beyond prior coursework and require self-directed learning, experimentation, and engineering judgment.

Interestingness:

I am extremely passionate about the automotive world, software engineering, and AI engineering. I personally own a 2022 Ford Escape Hybrid and have spent a large amount of time

learning everything I can about the vehicle, its software, and its powertrain. Building *EscapeAssist* is something that is personally meaningful to me, is something I am excited about, and will give me new technical challenges to overcome. This personal interest provides a motivation that will help me stay committed throughout the project and whatever difficulties arise.

Milestones, Tasks, and Schedule:

Total Required Effort: ~134 hours

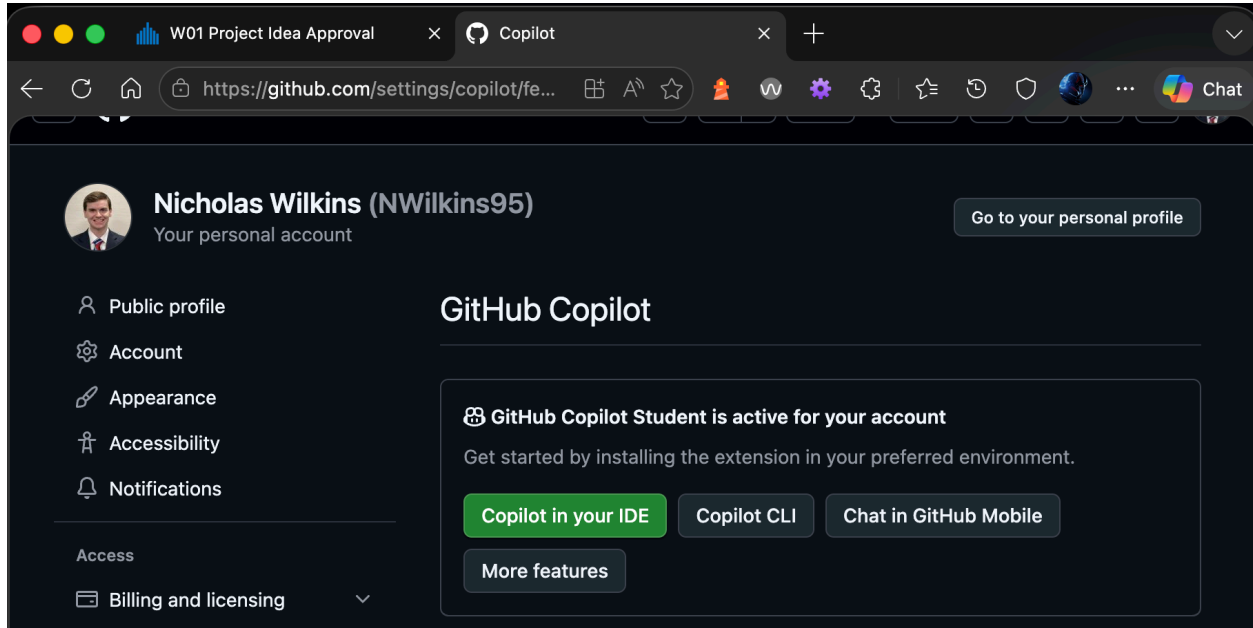
Week	Focus	Hours	Canvas Deliverables	Tasks	Milestone
Week 1	Research & Proposal	10	- W01 Status Report - W01 Project Idea Approval	Research RAG systems and Foundry capabilities; research Streamlit UI patterns; draft proposal; begin organizing manual sections	Proposal drafted + idea approved
Week 2	Proposal Completion & Foundry Setup	10	- W02 Status Report - W02 Proposal	Finalize proposal; create Foundry workspace; upload manual; run ingestion tests	Proposal submitted + Foundry environment ready
Week 3	Document Preprocessing Strategy	10	W03 Status Report	Split manual into sections; clean text; add metadata (section titles, categories); validate retrieval quality	Preprocessed manual ready
Week 4	Prompt Engineering & Retrieval Validation	12	- W04 Status Report - W04 Initial Requirements	Test prompt variations; evaluate grounding; identify failure cases; write Initial Requirements	Retrieval functional + Requirements submitted
Week 5	Streamlit UI Prototype	10	- W05 Status Report - W05 Technology Prototype	Build minimal chat UI; connect to Foundry endpoint; add chat history	End-to-end prototype working
Week 6	Evaluation Harness Design	10	W06 Status Report	Create golden question set; categorize questions; define scoring rubric; design judge LLM prompt	Evaluation harness design complete

Week 7	Judge LLM Integration	10	- W07 Status Report - W07 Requirements Specification	Implement judge scoring; store evaluation results; test hallucination detection; submit Requirements Spec	Judge LLM working + Requirements Spec submitted
Week 8	Dashboard Foundations	10	W08 Status Report (video)	Build accuracy charts; add summary statistics; create basic results table	Dashboard v1 complete
Week 9	Dashboard Completion	10	W09 Status Report	Add hallucination rate; add category-level metrics; add filtering/sorting; improve layout	Dashboard finalized
Week 10	Error Analysis & Iteration	10	W10 Status Report	Identify failure modes; improve preprocessing; improve prompts; re-run evaluation harness	System improved through evaluation cycles
Week 11	Retrieval Enhancements (Optional)	8	- W11 Status Report (video) - Instructor Meeting	Optional hybrid search; optional reranking; prepare for instructor meeting	Optional enhancements complete
Week 12	System Testing	8	W12 Status Report	End-to-end testing; UI refinement; fix edge cases; validate dashboard metrics	System stable
Week 13	Documentation	6	W13 Status Report	Write technical documentation; write user instructions; prepare demo script	Documentation complete
Week 14	Final Project & Reflection	6	- W14 Final Project - W14 Reflection - SPED Talk	Final polish; submit project; write reflection; deliver SPED Talk	Final project submitted

Resources:

- Microsoft Foundry (free tier)
- Azure/OpenAI embeddings (free tier or low cost)
- Streamlit
- Python

- Visual Studio Code
- 2022 Ford Escape Owner's Manual PDF
- Microsoft Copilot
- GitHub Copilot (free with GitHub Education)



Dependencies:

- No external permissions required (manual is public)
- Internet access
- Access to Microsoft Foundry
- Development on MacOS
- Python environment (Visual Studio Code)

Risks:

- Learning how to properly use Microsoft Foundry
- Designing an effective testing/evaluation harness
- Ensuring true and accurate retrieval
- Avoiding hallucinations
- Building meaningful dashboard metrics
- Managing scope to stay within 126 hours